

Ranking Transfers, Thresholds Don't: Unsupervised Test-Time Adaptation for Cross-Corpus Audio Deepfake Detection

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Abstract—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show that a fine-tuned self-supervised detector’s ranking (ROC-AUC) largely transfers cross-corpus even when its decision threshold does not calibrate, and put that asymmetry to work: where ranking survives, the model’s most-confident predictions on unlabeled target audio are reliable enough to self-train on. We exploit this with a stable unsupervised test-time adaptation (TTA) method: self-train on confident-ranked pseudo-labels and enforce channel-consistency, adapting only LayerNorm and the classifier head, with no target labels and no source data at test time. On a leave-one-corpus-out protocol over four targets (ASVspoof2019, a LibriSpeech-based TTS corpus, In-the-Wild, and an Arabic set), ten seeds each, EER improves on three (ASVspoof2019 5.03 $\rightarrow$ 3.47%, In-the-Wild 12.78 $\rightarrow$ 11.33%, Arabic 22.50 $\rightarrow$ 20.99%; two survive Holm correction across four per-target Wilcoxon tests) and is neutral on the fourth, which is also the target with the weakest source ranking (AUC 0.73). The method is not tied to our own detector: applied unmodified to four publicly released checkpoints we did not train, it improves In-the-Wild EER on all four (16.41 $\rightarrow$ 10.10% on the weakest, 5.04 $\rightarrow$ 4.08% on the strongest) by adapting 16,706 parameters (0.005%). Entropy-minimisation TTA (Tent) is instead unpredictably catastrophic (four of ten ASVspoof2019 seeds at or near chance, with a standard deviation as large as its mean), and a non-SSL backbone never reaches a ranking worth adapting. We beat a domain-adversarial baseline (DANN), which additionally sees unlabeled target audio during training, on two of four targets, and an aggregation-separation baseline (ASDG-style) on three of four. A proxy $\mathcal{A}$ -distance analysis shows domain separability does not decrease after adaptation, ruling out domain-invariance as the mechanism and supporting our ranking-calibration account instead.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, cross-corpus generalisation, self-supervised learning, EER.

I. Introduction

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [11].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high ROC-AUC while its accuracy at a fixed threshold

collapses—it still ranks clips correctly but its operating point is mis-calibrated. Because ranking transfers, its most-confident predictions on unlabeled target audio are trustworthy enough to adapt on, at test time, without any labels.

Our contributions: (i) an operational use of the ranking-calibration asymmetry—ranking transfers cross-corpus even when the threshold does not—as the precondition that makes unsupervised self-training viable; (ii) a simple, stable TTA method—confident-ranked pseudo-label self-training plus channel-consistency, adapting only LayerNorm and the head; (iii) a four-target, multilingual, ten-seed leave-one-corpus-out evaluation against seven comparison points (source-only, Tent, BatchNorm-only recalibration, self-training-only, a non-SSL backbone, DANN, and an ASDG-style aggregation-separation baseline), including the one target where the method does not help; (iv) a mechanistic analysis—proxy $\mathcal{A}$ -distance before/after adaptation—showing the gain does not come from erasing domain identity; and (v) evidence that the method acts as a label-free wrapper for third-party detectors, improving four publicly released checkpoints spanning two independent training corpora without retraining them.

Related work. Frozen or fine-tuned SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]; we additionally test whether the cross-corpus gap is an SSL question at all via a from-scratch, non-SSL backbone. Train-time domain generalisation (DG) methods align real and separate fake across source domains (aggregation-separation [5]) or run gradient-reversal domain-adversarial training over source and target jointly (DANN) [8]; both act at training time rather than at deployment. Entropy minimisation (Tent) [7] is the standard TTA recipe in vision but, as we show, is unstable here; batch-statistics recalibration [9] is a no-op for LayerNorm architectures. We borrow the proxy $\mathcal{A}$ -distance from Ben-David et al. [10] to test mechanism.

II. Method

Let x be a waveform with label $y \in \{0, 1\}$ (fake=1). A detector f_θ pools frame features into an embedding e and outputs $s_\theta(x) = \sigma(w^\top e + b) = P_\theta(y=1 \mid x)$: XLS-R-300M

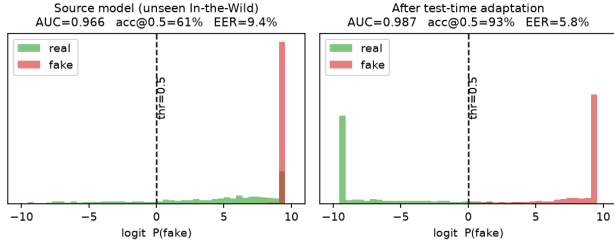


Fig. 1. Detector scores on unseen In-the-Wild audio. Left: the source model pushes many real clips past $\tau=0.5$ (accuracy well below AUC) yet still ranks most real below fake. Right: TTA shifts the real mass back across the threshold.

with only its top four transformer layers trainable, attentive temporal pooling, and a linear head. Its normalisation is LayerNorm, which carries no running batch statistics, so a batch-statistics-recalibration baseline is architecturally a no-op here and confidence-driven gradient adaptation is the minimal mechanism that can move these layers at all.

The ranking-calibration gap. Ranking quality $AUC = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free and invariant to reparametrisation of s_θ ; accuracy at a fixed threshold τ is not. Empirically, AUC degrades far less than accuracy at $\tau=0.5$ under source $\rightarrow$ target shift (Fig. 1). We therefore trust score order: the extreme quantiles of unlabeled target scores are higher-precision pseudo-labels than a fixed threshold, even when s_θ is mis-calibrated. The fixed-quantile split Q_q vs. Q_{1-q} implicitly assumes both a reliable source ranking and roughly balanced target classes; we scope the method to that regime, and Sec. IV and Sec. V show what happens when each assumption breaks.

Test-time adaptation. Each epoch, we assign confident pseudo-labels from the q - and $(1-q)$ -quantiles of target scores, ignoring the middle; self-train with cross-entropy on this confident set ($\mathcal{L}_{ST}$); and enforce prediction invariance under a stochastic channel perturbation $a(\cdot)$ (gain+noise) via an MSE consistency term $\mathcal{L}_{cons}$ between clean and perturbed views (stop-gradient on the clean target). The objective is

$$\min_{\{\gamma, \beta\}, \phi} \mathcal{L}_{ST} + \lambda \mathcal{L}_{cons}, \quad \lambda=0.3, q=0.3, E=4, \quad (1)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned layers and the head ϕ ; the rest stays frozen. Restricting the trainable set and using only confident tails prevents the degenerate collapse to which entropy minimisation is prone.

Domain divergence. To test whether adaptation works by making source and target indistinguishable in embedding space (as adversarial DG methods explicitly optimise for), we compute a proxy $\mathcal{A}$ -distance $\hat{d}_A = 2(1-2\epsilon_h)$ following Ben-David et al. [10], where ϵ_h is a linear domain classifier’s held-out error on frozen embeddings, before and after adaptation.

TABLE I
Cross-corpus EER % / AUC, mean over 10 seeds
(leave-one-corpus-out, transductive). Best of {Source, Tent,
ST-only, Ours} per column in bold.

Method	ASVspoof2019	LibriSpeech-TTS	In-the-Wild	Arabic
Source-only	5.03 / .989	33.54 / .727	12.78 / .945	22.50 / .843
Tent [7]	23.47 $\pm$ 21.96 / .776	35.44 $\pm$ 3.20 / .674	27.34 $\pm$ 11.21 / .735	47.35 $\pm$ 5.03 / .527
Self-training only	4.20 / .987	33.39 / .709	11.55 / .950	22.19 / .846
Ours (TTA)	3.47 / .994	33.74 / .727	11.33 / .958	20.99 / .866
Oracle (target-supervised)	0.89 / .999	13.93 / .939	4.42 / .992	9.50 / .968

III. Experimental Setup

Protocol. Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA [12]; D2, a corpus pairing LibriSpeech bona fide speech with fakes from eight open TTS/voice-cloning systems (NaturalSpeech3, FlashSpeech, VALL-E, XTTS, PromptTTS2, VoiceBox, SeedTTS and a commercial API); In-the-Wild [11]; and the Arabic Audio Deepfake set (ArAD) [14]. For each held-out target, the source pool merges the other three corpora with a 38-language MLAAD subset [13] (fake-only) for generator/language diversity; held-out target audio, including its MLAAD-overlapping language content, is never in any source pool, and six MLAAD languages are held out of every source pool entirely to test genuine unseen-language generalisation.

Comparison points. Source-only (no adaptation); Tent [7] (entropy minimisation, same trainable set as ours); BN-only (AdaBN-style recalibration [9], no gradient step); self-training-only (ours with $\lambda=0$); Oracle (fine-tuned on target labels, a supervised upper bound); RawNet2Lite (a from-scratch, non-SSL backbone in the RawNet2 family [4], trained identically, to isolate whether robustness comes from the SSL front-end); DANN [8], trained to convergence with the held-out target’s unlabeled inputs drawn every step through the gradient-reversal branch—strictly more privileged access than our method has—and an ASDG-style aggregation-separation baseline [5], which aligns across the source domains only and never sees target audio.

Implementation. 16 kHz audio, 3 s crops, top-4-layer XLS-R fine-tuning, Adam $lr 2 \times 10^{-4}$, batch 32, source trained 8 epochs. Adaptation reuses the same batch size and crop length with $lr 1 \times 10^{-4}$ for $E=4$ passes; the perturbation $a(\cdot)$ is a per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus additive Gaussian noise ($\sigma=0.005$), and the head is carried over from the source model rather than re-initialised. Every number in Tables II and I, and every significance test, is over ten seeds per target for ours and for each baseline in those tables. The third-party-checkpoint, proxy $\mathcal{A}$ -distance and ASVspoof2021-DF results are smaller studies and we give their run counts where they appear. Code and manifests: [URL withheld for review].

IV. Results

Table I is the main result. Our method wins on three of four targets: ASVspoof2019 ($5.03 \rightarrow 3.47\%$, 10/10

seeds), Arabic (22.50 $\rightarrow$ 20.99%, 10/10), In-the-Wild (12.78 $\rightarrow$ 11.33%, 8/10); on LibriSpeech-TTS it is EER-neutral (33.54 $\rightarrow$ 33.74%, improved in only 5/10 seeds), and that is also the target with the lowest source AUC (0.73 vs. $\geq$ 0.84 elsewhere). The ordering is consistent with our ranking-precondition account, but the association is between corpora, not within them: the four target means give $r=+0.86$ ($n=4$, $p=0.14$), while pooling all 40 (target, seed) points gives $r=+0.47$ ($p=0.002$)—an ecological artefact of four clusters, since within each target the sign is reversed or absent ($r=-0.77$, -0.64 , -0.64 , $+0.12$; Fig. 2). We therefore report source AUC as a four-corpus descriptive ordering, not a validated deployment rule.

Significance. Ten seeds per target make a per-target Wilcoxon test possible at all ($n=5$ has a combinatorial floor of 0.0625; $n=10$ reaches 0.002). Against source-only, ours is significant on ASVspoof2019 and Arabic ($p=0.0020$ each) and In-the-Wild ($p=0.0293$), not on LibriSpeech-TTS ($p=0.23$). Holm correction across the four keeps ASVspoof2019 and Arabic ($p_{\text{adj}}=0.0078$) but not In-the-Wild (0.0586), which we therefore call nominal. A 10k-resample bootstrap on the ten per-seed EER differences agrees, excluding zero on ASVspoof2019 [+1.12, +2.02], In-the-Wild [+0.43, +2.33] and Arabic [+1.10, +1.88], and including it on LibriSpeech-TTS [-0.49, +0.09]. Pooled over all 40 pairs the method beats source-only ($p=2.4 \times 10^{-5}$), Tent (1.5×10^{-7}), self-training-only (0.0056) and both DG baselines ($p<10^{-4}$, 30/40 each); these pooled figures treat clustered pairs as independent and are descriptive only.

Tent is not merely unstable, it is unpredictably catastrophic. Its ASVspoof2019 per-seed EER is 4.16, 3.91, 3.04, 49.52, 3.42, 49.44, 49.69, 42.11, 25.31, 4.12%—five seeds close to source-only, four at or near chance and one midway, with no warning sign; the standard deviation (21.96) is comparable to the mean (23.47). On Arabic it is a consistent failure instead ($47.35 \pm 5.03\%$, chance-level in 9/10 seeds). Our method shows no comparable collapse on any target (largest std 2.01). The failure is not specific to our source model, and is a ranking failure first: run unmodified on two of the four public checkpoints below (the strongest and the weakest), Tent collapses to chance on one (16.41 $\rightarrow$ 49.80% EER, AUC .918 $\rightarrow$.502) [16] and leaves EER near-unchanged on the other (5.04 $\rightarrow$ 4.52%) while still degrading AUC (.986 $\rightarrow$.979) [15]—an EER collapse appears only where the model had AUC headroom to lose.

Transfer to third-party detectors. Every result above adapts our own source model, so we apply the published configuration unmodified to four released detectors we did not train—one WaveFake-trained [15], three ASVspoof2019-trained [16] XLS-R+AASIST checkpoints, none having seen In-the-Wild. EER and AUC improve on all four—5.04 $\rightarrow$ 4.08, 9.59 $\rightarrow$ 9.06, 8.41 $\rightarrow$ 7.88 and 16.41 $\rightarrow$ 10.10% EER, with AUC .986 $\rightarrow$.990, .966 $\rightarrow$.970,

TABLE II

Test-time adaptation vs. two train-time domain-generalisation baselines, mean EER % / AUC over the same 10 seeds for every method. Best per row in bold. Ours wins 30/40 (target, seed) pairs against each; per target we beat ASDG on three of four and DANN on two of four, and ASDG is better on LibriSpeech-TTS. Only DANN sees target audio during training. ASDG ran on different hardware, so its wall-clock is not compared.

Target	Source	DANN [8]	ASDG [5]	Ours
ASVspoof2019	5.03 / .989	7.35 / .966	4.98 / .987	3.47 / .994
LibriSpeech-TTS	33.54 / .727	34.43 / .717	32.64 / .749	33.74 / .727
In-the-Wild	12.78 / .945	13.34 / .924	17.33 / .891	11.33 / .958
Arabic	22.50 / .843	28.53 / .759	27.41 / .798	20.99 / .866

.970 $\rightarrow$.973 and .918 $\rightarrow$.961, over all 31,779 clips—while adapting 16,706 parameters (0.005%). Three of the four gains are modest (0.52–0.96 EER); the large one (6.31) belongs to the checkpoint arriving worst-calibrated (accuracy 39.0% at threshold 0.5 despite AUC .918)—the regime the precondition predicts. It is not memorisation of the adaptation pool: adapting on half of In-the-Wild and evaluating on the disjoint half gives 4.08% vs. 4.06% transductive, and 10.21% vs. 10.10%. The three ASVspoof2019 checkpoints, identical but for training seed, span 8.41–16.41% source EER—single-checkpoint cross-corpus figures, ours included, carry a large and usually unreported error bar.

Backbone and prior DG baselines. RawNet2Lite (no SSL pretraining) stays near chance on every target (AUC 0.40–0.65), confirming cross-corpus viability starts from SSL pretraining, not the TTA step; retraining it for 40 epochs with stronger regularisation does not close the gap, ruling out undertraining as the explanation. Both train-time baselines are beaten pooled over all 40 (target, seed) pairs (30/40 each), but at ten seeds the per-target picture is narrower than a three-seed comparison suggested, and differs between them. Against DANN—which additionally samples the held-out target’s unlabeled audio at every training step—we win on ASVspoof2019 (9/10, $p=0.0098$) and Arabic (10/10, $p=0.0020$) and tie on In-the-Wild (5/10) and LibriSpeech-TTS (6/10). Against ASDG—which trains on source domains only—we win on ASVspoof2019 (8/10, $p=0.0098$), In-the-Wild (10/10, $p=0.0020$) and Arabic (10/10, $p=0.0020$), with ASDG better on LibriSpeech-TTS (32.64% vs. our 33.74%, 2/10, $p=0.16$). DANN is markedly worse on Arabic, about $3\times$ our adaptation cost (20.1 vs. 6.9 min per target-seed over ten seeds), and the less seed-stable of the two (ASVspoof2019 EER ranges 3.23–14.85%); Tent remains the least stable method overall by a wide margin. ASDG beats DANN on three of four targets but plain source-only on two (Table II). Neither is a reliable route to cross-corpus robustness here.

Mechanism. An inductive check (adapt on one half of the target, evaluate on the disjoint half) confirms the gain is not transductive memorisation. On one seed per target, the proxy $\mathcal{A}$ -distance and a linear domain-

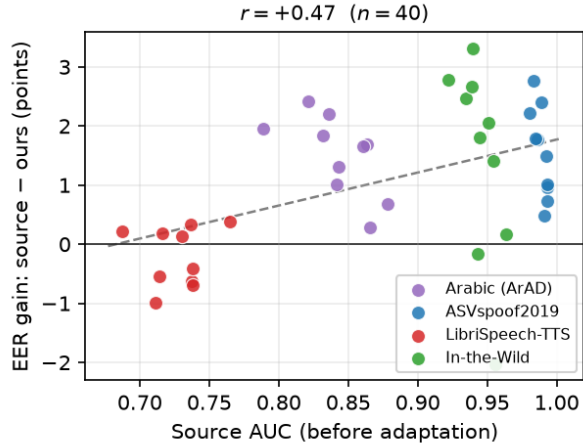


Fig. 2. EER gain (source—ours) vs. source AUC before adaptation, one point per (target, seed), all 40 runs. The association is between corpora, not within them: the four target means give $r=+0.86$ ($n=4$, $p=0.14$) and the pooled 40-point fit gives $r=+0.47$, but within each target the sign is reversed or absent ($r=-0.77$, -0.64 , -0.64 , $+0.12$). LibriSpeech-TTS (red, lowest AUC) is the only target with a negative mean gain.

origin probe both increase slightly after adaptation rather than decreasing (e.g. Arabic $\hat{d}_A$: 1.545 $\rightarrow$ 1.606)—small shifts, but in the opposite direction to what domain-invariance predicts. Adaptation re-shapes the decision boundary using rank-reliable pseudo-labels rather than erasing domain identity, consistent with DANN underperforming despite explicitly optimising that quantity. A 38-language MLAAD breakdown shows fake-clip recall is uniform across seen and unseen languages (93.9% vs. 93.3%).

V. Discussion and Limitations

The method is transductive (unlabeled target inputs at adaptation time, the standard TTA setting); an inductive check reproduces every result including the LibriSpeech-TTS null. We report per-target, per-seed numbers throughout, so the limitations below can be read against the evidence:

- Needs an already-reliable source ranking—and we cannot say how much of that is SSL. A from-scratch or low-AUC starting point is not a regime we can rescue (Fig. 2). Our non-SSL backbone never reaches a ranking worth adapting (AUC 0.40–0.65), so it fails the precondition before TTA becomes applicable and we did not run adaptation on it; all four third-party checkpoints the wrapper improves are themselves SSL front-ends.
- Class balance is a second precondition, alongside AUC. On the official ASVspoof2021-DF protocol (train on 2019-LA alone; 97.2%-spoof eval pool, of which we score 75%), source-only reaches 26.33% EER and TTA actively harms (42.46%). The cause is structural: the fixed-quantile rule labels the bottom q

of the pool “confident real” whatever the true prior, so under severe skew that bucket fills with mislabelled spoof and self-training teaches the model that much of the fake class is real. Two label-free corrections fail—a prevalence-weighted budget helps under one source recipe (32.58%) but not another (42.46%), and a curriculum over q changes nothing (42.50%)—both still worse than not adapting.

- Not a state-of-the-art detector. Our source model is an XLS-R encoder with a light attentive head trained on a leave-one-corpus-out pool, not a system tuned for any single benchmark: on 2021-DF, published SSL-plus-AASIST systems reach 8–9% EER and specialised AASIST-class systems 2–3%, against our 26.33% source-only. The claim here is the relative gain from adaptation on a fixed backbone under an identical protocol, not a competitive absolute result; the mechanism would have to be re-verified on a stronger source model.
- Fixed hyperparameters. q, λ, E are chosen without target labels and are not per-target optimal; a label-free schedule for λ is predictable from tail separation but pays nothing at our scale (fixed λ is within 0.24 EER of a per-point oracle), so λ stays constant.
- Scale. Four targets, ten seeds. Going from three to ten seeds made the per-target tests possible and confirmed Tent’s collapse as routine, but shrank our DG margin to two per-target wins against DANN and three against ASDG. Seeds do not buy corpora: the AUC ordering rests on four points ($n=4$, $p=0.14$), so the precondition is a hypothesis our data are consistent with, not one they test.

VI. Conclusion

Cross-corpus audio deepfake detection fails not because ranking is lost but because the operating point is miscalibrated. A simple unsupervised test-time adaptation exploiting this improves generalisation on three of four leave-one-corpus-out targets—two surviving Holm correction—and is neutral on the fourth, the one with the weakest source ranking. Whether source AUC predicts the gain is suggested by our four corpora but not established by them ($r=+0.86$, $n=4$, $p=0.14$); settling it needs more corpora, not more seeds. Entropy-minimisation TTA is unpredictably catastrophic, a non-SSL backbone never reaches an adaptable ranking, and two train-time DG baselines are beaten pooled over all 40 pairs—one despite seeing unlabeled target audio during training, at $3\times$ our cost. A proxy divergence analysis puts the mechanism at re-calibration, not domain-invariance; needing neither labels nor retraining, the method also improves four released SSL detectors—a wrapper, not a system.

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